Behaviour-based Malware Detection and Classification System

Cybercrime & IT Security Final Year Project Proposal

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Cybersecurity Problem and Motivation

Ransomware is a type of malicious software that cybercriminals use to encrypt and deny access to a victim's data until a ransom payment is made (Microsoft, 2025). The effects of a ransomware attack can be devastating, from victims losing money due to paying the ransom, to organisations being unable to function as normal.

For my project, I intend to develop a behaviour-based malware detection and classification system focused on ransomware. Traditional signature-based detection relies on identifying known characteristics of previously identified malware. However, ransomware can be modified to evade signature-based detection, which reduces the effectiveness of these techniques. Behaviour-based detection provides an alternative approach by identifying suspicious activity as it occurs, rather than relying solely on a known signature. This system could significantly improve early detection and potentially stop the ransomware attack before full disk encryption occurs.

Aim and objectives

The aim of this project is to develop and evaluate a behaviour-based malware detection and classification system, which can identify ransomware activity and alert a specified system when such activity is detected.

To achieve this aim, the objectives will be as such:

1. Research existing ransomware detection and classification techniques, with a focus on behaviour-based approaches
2. Identify the behavioural characteristics and file system activity typically associated with ransomware.
3. Design and develop a program which can detect ransomware-like activity.
4. Test the detection program using legitimate and controlled ransomware activity.
5. Investigate whether behavioural characteristics can be used to classify different families or types of malware.
6. Evaluate the performance of the system using measures such as detection accuracy, detection time and false-positive rate.

Research question: To what extent can behaviour-based analysis be used to detect and classify ransomware activity in a controlled environment?

Technical Approach:

The proposed system will be developed as a behaviour-based malware detection program with ransomware as the main focus. The system will be developed and tested in a secure and controlled virtual environment to ensure that no malware can impact any external systems.

A program will be developed to imitate file-system behaviour that reflects how ransomware modifies an affected system. A detection program will then be developed to monitor activity within the test environment and collect behavioural and file system indicators that may be associated with malicious activity. This will include indicators such as rapid file modification, file extension changes, and unusual access patterns.

The project will also investigate whether the behavioural characteristics identified during the detection process can be used to classify different families or types of malware. Ransomware will remain the primary focus of the project, but the system may be extended to investigate other types of malware if sufficient time is available.

Code development will be conducted primarily in Python.

Testing will include both legitimate system activity and controlled ransomware-like activity within the isolated environment. The results will assess how effectively the system can differentiate between legitimate file-system activity and malicious ransomware activity and used to improve the system to reduce the false positive rate.

Cybersecurity Contribution

This project aims to contribute to the understanding of how behavioural and file system analysis can be applied to defensive security operations. Rather than relying on previously known signatures, this project will explore whether observable file system activity can provide useful information about unknown or modified malware.

The completed prototype will provide a practical example of how behavioural analysis can be incorporated into existing endpoint security and malware detection solutions.

Evaluation

The system will be evaluated by testing it against a range of legitimate file-system activity and controlled ransomware activity within the isolated environment. This will show the system's ability to distinguish between normal and potentially malicious behaviour.

The effectiveness of this system will be measured using metrics such as false positive rate, detection time and detection accuracy. The results of this will be used to identify any weaknesses in the system, and what needs to be revised and refined.

Scope and Feasibility

The initial scope of this project will be to detect ransomware activity and ransomware-like behaviour through behavioural and file-system analysis within a controlled virtual environment. The core deliverable will be a functional program capable of identifying malware activity and generating an alert when suspicious behaviour is detected.

The project will initially stay focused on detecting specifically ransomware activity to ensure that it remains achievable within the available timeframe. The detection and classification of different types of malware may be investigated as an extension to the core functionality.

This project is considered feasible due to the availability of resources such as open-source malware sandboxes, virtual machine images, other existing cybersecurity tools and Python for development. Using an isolated virtual machine ensures the testing can be carried out safely without affecting any external systems. The system will be developed incrementally, with the core detection functionality being completed before additional features are considered.

Project Proposal Screencast:

Behaviour-Based malware detection and classification.mp4

References:

Microsoft (2025) *What is ransomware?* Microsoft Security.

https://www.microsoft.com/en-ie/security/business/security-101/what-is-ransomware

Enyindah, Promise & Okon, Daniel. (2026). Ransomware Detection Using Behavioral Analysis and Machine Learning: A Comprehensive Review. International Journal of Science Engineering and Technology. 14. https://www.researchgate.net/publication/408647863_Ransomware_Detection_Using_Behavioral_Analysis_and_Machine_Learning_A_Comprehensive_Review